Master's Thesis Summary

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Overview. My master's thesis investigates how atmospheric amines are emitted, transformed, removed, and incorporated into new particle formation in a global Earth system model. The work has two connected stages. The first develops a **global source and sink framework** for methylamine (MMA), dimethylamine (DMA), and trimethylamine (TMA) in CESM/IMPACT and evaluates the simulations against observations. The second moves from model application to **source code development** by implementing a multi mechanism **new particle formation** framework in CESM 1.2.2 and coupling amine and sulfuric acid chemistry with aerosol microphysics.

Stage I - Global amine emissions and atmospheric cycling. I developed a global gridded emission inventory for MMA, DMA, and TMA by combining six anthropogenic sectors with marine biological emissions. Anthropogenic emissions were estimated using source specific amine to ammonia ratios applied to EDGAR ammonia emissions, while marine emissions were calculated from air sea exchange using global meteorological and oceanographic data, including sea surface temperature, wind speed, salinity, pH, and modeled ammonia. I then implemented the three amines as atmospheric tracers in CESM 1.2.2 with the IMPACT aerosol module and added the major removal processes of OH oxidation, wet deposition, dry deposition, and heterogeneous aerosol uptake. The model was run globally at 1.9° x 2.5° resolution with 30 vertical layers and evaluated against observations from 27 sites covering urban, agricultural, forest, coastal, marine, and polar environments.

To use the model as an analytical tool rather than only to generate concentration fields, I designed seven source attribution experiments that isolated agriculture, chemical industry, residential activity, transportation, biomass burning, other industry, and ocean sources. I combined these experiments with a source sink budget analysis to quantify how emissions, oxidation, deposition, and aerosol uptake control global and regional burdens. I also calculated spatially varying atmospheric lifetimes, examined seasonal and hemispheric behavior, and used sensitivity simulations to test uncertain uptake parameters and marine exchange assumptions. Comparison with observations was used not only as performance evaluation but also to diagnose whether model biases were more consistent with emission uncertainty, missing chemical processes, or the representativeness limits of coarse global grids.

These analyses showed that agriculture dominates MMA and DMA emissions, while marine biological processes account for about 98% of global TMA emissions. Gas phase oxidation is the largest global sink, and simulated lifetimes of roughly 3-5 hours explain why concentrations remain closely tied to nearby sources. Adding the ocean source substantially improved TMA simulations in marine influenced regions, whereas persistent underestimation across all three species indicated unresolved emissions and secondary formation pathways. This stage resulted in my first author paper in the **Journal of Geophysical Research: Atmospheres (2026).**

Stage II - CESM source code development for new particle formation. The second stage extends the thesis from atmospheric cycling to aerosol formation. I built and implemented a multi mechanism new particle formation framework in CESM 1.2.2 by adapting the CMAN framework from E3SM to the different CESM/IMPACT architecture. This required writing Fortran interfaces between sulfur chemistry and aerosol microphysics, mapping chemical species and tracers, reconciling units and array structures, and routing newly formed particle number and mass into the appropriate sulfate and organic aerosol modes. I integrated amine and sulfuric acid nucleation, precursor consumption, particle number and mass allocation, coagulation, and modal transfer, and extended the framework to represent MMA, DMA, and TMA nucleation using ACDC based parameterizations and literature constraints.

The complete framework now compiles and runs successfully. My current work focuses on validation and refinement because some aerosol responses differ from the expected physical behavior. I use source code diagnostics and controlled sensitivity simulations to trace mass and number conservation, unit conversion, precursor depletion, aerosol mode indexing, coagulation, and process routing through the coupled model. This debugging process has required following variables across multiple Fortran modules and distinguishing numerical or structural errors from genuine physical responses, giving me practical experience with model architecture as well as atmospheric aerosol processes.

Overall significance. Together, the two stages span the full modeling workflow from emission inventory construction and global simulation to observational evaluation, sensitivity analysis, aerosol microphysics, HPC computing, and direct source code development. Because the work repeatedly uses ammonia and amine emissions, chemistry, and observational datasets, it has also given me substantial practical familiarity with reduced nitrogen species and their representation in global models. More broadly, the thesis has trained me to identify missing or simplified atmospheric processes, implement them in model code, design numerical experiments, diagnose unexpected behavior, and test simulations against observations. My main research interest is to carry this process based model development forward from aerosol formation to aerosol cloud interactions, where the representation of aerosol and cloud microphysics remains a leading source of uncertainty in climate models.